Event ends in 1 day 4 hours 8 minutes.

Prerequisites 📋

Important
- Use Kiro **CLI** for this section
- Complete Setting Up Kiro

Situation 🔍

As a cloud engineer, you've been tasked by the management team to help with cost optimization. You noticed in Cost Explorer that there was a high volume of data going through the NAT gateway, and you need to investigate this issue. We have default VPC flow logs set up 📊

Following the Making Sense of VPC Traffic with Kiro CLI 🔄 section, we were able to see the individual traffic going in and out of the NAT Gateway, but the question is which network flow exactly, which EC2 instance is talking to which service. To dive deeper, VPC flow logs support custom fields 🔧

▶ **Address Fields in VPC Flow Logs**

---

1. Use Kiro to Configure a Custom VPC Flow Log 📝

1. In your VS Code Server environment, create a new folder to separate context from other files inside VS Code Server:

```
cd /Workshop
mkdir deep-vpc-analysis
cd deep-vpc-analysis
```

2. Within the VS Code Server environment, start Kiro CLI by running:

```
kiro-cli chat
```

3. Ask Kiro CLI to create a VPC flow log for the VPC 📄:

```
Create a new log group and VPC flow log for VPC name 'sampleapp-VPC' in us-east-1
Capture at 1-minute intervals and store it in CloudWatch logs.
Use a log format that allows logs to see traffic going through the NAT gateway, as well as which AWS service.
Output the flow log details and VPC CIDR to a markdown file custom-flowlog.md
```

(Optional) You can also use this prompt in your environment by changing the VPC ID and AWS Region 🔄

4. Once complete, Kiro will report that the job is done ✅:

```
 :
 ● Completed in 0.841s
Great! I've successfully created the VPC flow log with the following configuration:
```

custom-flowlog.md ×

deep-vpc-analysis > custom-flowlog.md > # Custom VPC Flow Log Configuration > ## Flow Log Details

```
# Custom VPC Flow Log Configuration
## Flow Log Details

- **Flow Log ID**: fl-01ca558fec4bd5776
- **VPC ID**: vpc-vpcid
- **Status**: ACTIVE
- **Creation Time**: 2025-06-16T04:34:16.576000+00:00
- **Region**: us-east-1
- **Traffic Type**: ALL
- **Max Aggregation Interval**: 60 seconds (1 minute)

## Log Destination

- **Destination Type**: CloudWatch Logs
- **Log Group Name**: /aws/vpc/flowlogs/vpc-vpcid
- **Log Group ARN**: arn:aws:logs:us-east-1:accid:log-group:/aws/vpc/flowlogs/vpc-vpcid
- **IAM Role ARN**: arn:aws:iam::accid:role/VPCFlowLogsRole
- **Delivery Status**: SUCCESS

## Custom Log Format

The flow log uses a custom format that includes fields for NAT gateway traffic and AWS service identification:

```
${version} ${account-id} ${interface-id} ${srcaddr} ${dstaddr} ${srcport} ${dstport} ${protocol} ${packets} ${bytes} ${start} ${end} ${action} ${log-status} ${vpc-id} ${subnet-id} ${instance-id} ${tcp-flags} ${type} ${pkt-srcaddr} ${pkt-dstaddr} ${region} ${az-id} ${sublocation-type} ${sublocation-id} ${pkt-src-aws-service} ${pkt-dst-aws-service} ${flow-direction} ${traffic-path}
```

---

2. Use Kiro to Tail Logs in CloudWatch 📋

1. Continue the Kiro session. If you accidentally close the session, you can resume with:

```
kiro-cli chat --resume
```

2. Ask Kiro CLI to tail the log group that we just created 👀:

```
tail the vpc flowlog
```

3. Wait for VPC flow logs to start flowing in ⏳
4. Clear Kiro chat session 🚪:

```
/exit
```

---

3. Generate Network Traffic Visualization Using Kiro 📊

1. Let's ask Kiro CLI to identify the top 20 IP traffic sources going through the NAT gateway, focusing on pkt-srcaddr and pkt-dstaddr 🔎:

```
What is the top 20 traffic by byte transfer
Focus on pkt-srcaddr, pkt-dstaddr, pkt-src-aws-service, pkt-dst-aws-service
Ignore srcaddr, dstaddr
Query traffic from VPC flow log details in custom-flowlog.md
Make sure to check log format to see the fields
Output to traffic_table.csv
```

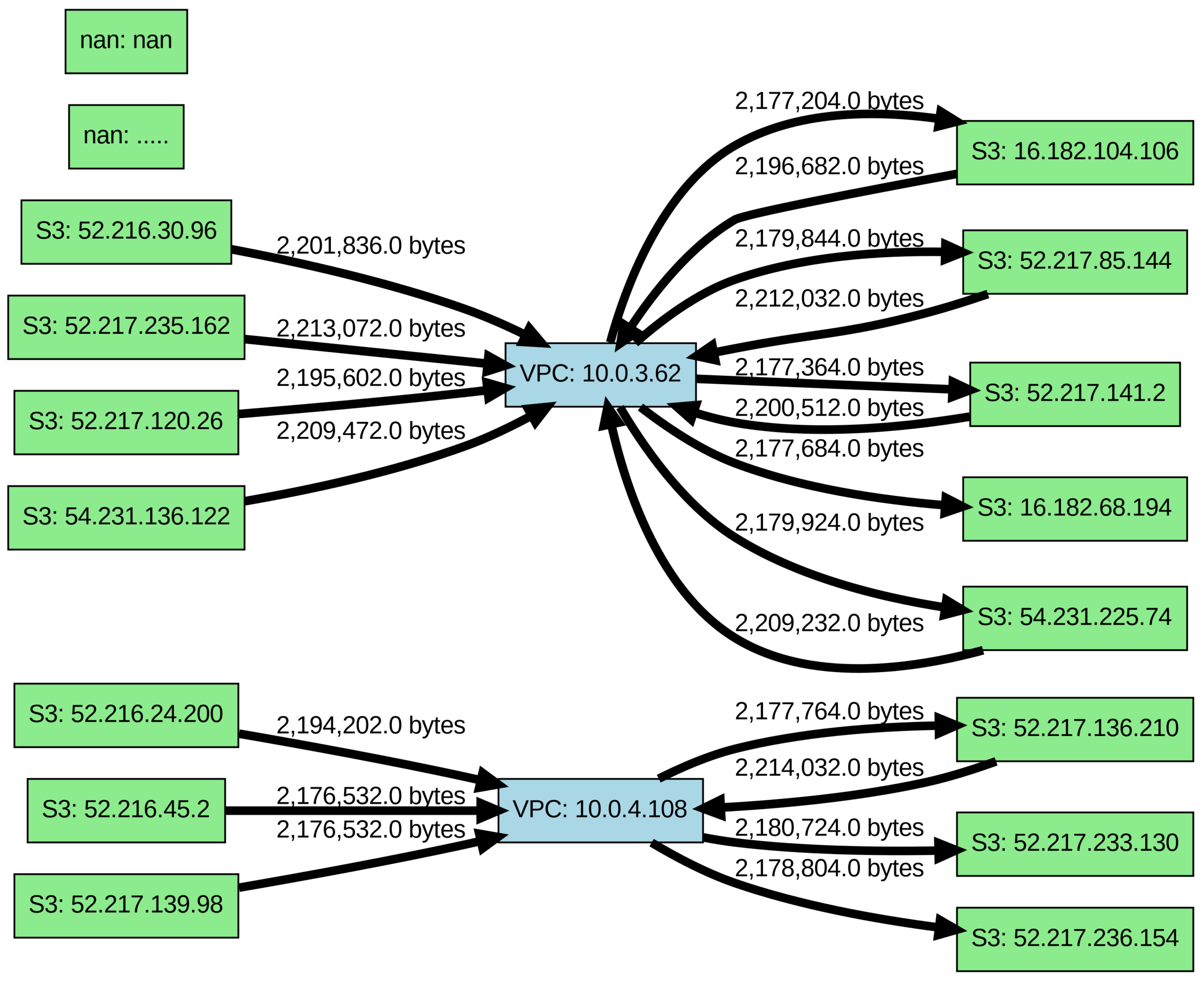
custom-flowlog.md | traffic_table.csv ×

deep-vpc-analysis > traffic_table.csv

```
pkt_srcaddr,pkt_dstaddr,pkt_src_aws_service,pkt_dst_aws_service,bytesTransferred
52.217.136.210,10.0.4.108,S3,–,2214032
52.217.235.162,10.0.3.62,S3,–,2213072
52.217.85.144,10.0.3.62,S3,–,2212032
54.231.136.122,10.0.3.62,S3,–,2209472
.....
```

2. Visualize the data going through the NAT Gateway 🎨:

```
Using traffic_table.csv, create a network visualization diagram.
Node is IP address, with label as IP
Label:
* VPC
* AWS services (Service name: S3, DynamoDB, etc.)
* Internet
Edge is a directed graph between pkt-srcaddr and pkt-dstaddr
Edge weight based on total data transfer volume
Use networkx and graphviz to visualize.
```



---

3. **Conclusion** 💡 - Using visualization, you can track Virtual Private Cloud (VPC) traffic patterns by analyzing both origin and destination points. You are now able to drill down into detailed traffic flows and optimize your network accordingly. Our analysis shows the top 20 traffic flows from the VPC to Simple Storage Service (S3). To optimize this application, we recommend creating an S3 gateway endpoint, which will both reduce data transfer costs and ensure private network flow 💰

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